

Repsol Electra 3

Paraffinic dielectric oil — Uninhibited

PRODUCT DATA SHEET

Description

ELECTRA oils are light, paraffinic oils designed specifically for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc. In general, they can be used in any kind of electrical appliance requiring an oil bath which acts as a dielectric or coolant. They are manufactured from highly-refined treated base oils which ensure the lack of any solid, polar or water compounds. The hydrogenation process guarantees that they do not contain any corrosive sulphur, which has been proven to be extremely damaging to transformers.

Classification and standards

ELECTRA 3 is an uninhibited oil that complies with the IEC 60296 and ASTM D3487 Type I standards.

Technical specifications

Values according to IEC 60296: 2012 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		Visual		Bright and clear	
Density at 20 °C	g/cm3	ISO 12185		0.895	0.84
Viscosity at 40 °C	mm2/s	ASTM D445		12	10.3
Viscosity at 100 °C	mm2/s	ASTM D445		3	2.7
Viscosity at -30 °C	mm2/s	ASTM D445		1800	807
Pour point	°C	ASTM D97		-40	-48
Interfacial tension	dynes/cm	ASTM D971	40		42
Chemical properties					
Acidity	mg KOH/g	ASTM D974		0.01	<0.01
Potentially corrosive sulphur		IEC 62535		Non-corrosive	
Corrosive sulphur		DIN 51353		Non-corrosive	
Corrosive sulphur		ASTM D1275B		Non-corrosive	
Antioxidant additive	% weight	IEC 60666		Exempt	
Water content	mg/kg	IEC 60814		30	15
Furfural content	mg/kg	IEC 61198		0.05	<0.05
Electrical properties					
Dielectric loss factor at 90 °C		IEC 60247		0.005	0.00074
Interfacial tension	mN/m	UNE 21320	40		42
Breakdown voltage, untreated	kV	IEC 60156	30		35

Properties	Units	Method	Min.	Max.	Typical value
Breakdown voltage, treated	kV	IEC 60156	70		>70
Oxidation stability — IEC 61125C (164 h / 120 °C)					
Total acidity	mg KOH/g	IEC 61125C		1.2	0.39
Sludge	% weight	IEC 61125C		0.8	0.084
Dielectric loss factor at 90 °C		IEC 61125C		0.5	0.0707
Health, safety and the environment					
Flash point	°C	ASTM D93	160		170
PCA content	% weight	IP 346		3	0.8
PCB content	% weight	IEC 61619		Exempt	

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.